

The Economics of Cognitive Bandwidth

Wetware Stack Mini Essay #1 by Travis R. McKinzie

Master Prompt: Meditations on Man as Machine

The Hook: For millennia, Stoics and Buddhists have taught that to achieve peace, we must detach from our immediate emotional reactions and observe our thoughts without judgment. Historically, this has been framed as a spiritual achievement or an alignment with the universe. Marcus Aurelius and the Buddha were blind engineers who discovered how to hack the human nervous system. They didn't find the soul; they found the manual for the fleshy computer.

The Core Thesis I am Defending: Arguing that ancient mindfulness and Stoic philosophy are not spiritual disciplines, but rather rigorous, metabolically expensive **neuro-interventions** designed to protect the brain's hardware from the ravages of human life in civilized societies.

1. The Evolutionary Context (The Origin & The Flaw) *How the hardware was designed and why it fails today.*

- [[Evolutionary Mismatch]]
- [[EEA (Environment of Evolutionary Adaptedness)]]
- [[Kin Selection]]
- [[Social Hierarchy (Biology)]]

2. Macro-Anatomy & Systems (The Hardware) *The physical structures and wiring of the brain.*

- [[The Central Nervous System (CNS)]]
- [[The Autonomic Nervous System (ANS)]]
- [[The Prefrontal Cortex (PFC)]]
- [[The Amygdala]]
- [[The Limbic System]]

3. The Chemical Environment (The Weather & The Fuel) *The messengers that alter the sensitivity of the hardware.*

- [[Hormone]] (Cortisol, Testosterone)
- [[Dopamine (Reward Prediction Error)]]

4. Micro-Anatomy & Synaptic Mechanics (The Computations) *How individual neurons communicate and alter their connections.*

- [[Neural Network]]
- [[Pre vs Post Synaptic Changes]]
- [[Nonlinear Excitation]]
- [[Long Term Potentiation (LTP)]]
- [[Long Term Depression (LTD)]]
- [[Myelination]]

5. Biological Adaptation & Wear-and-Tear (The Ledger) *The long-term physical consequences of using the hardware.*

- [[Neuroplasticity]] (Functional & Structural)
- [[Neurogenesis]]
- [[Sustained Stress]]
- [[Allostatic Load]]

6. The Cognitive Output (The Interface) *The algorithmic result of the hardware in action.*

- [[System 1 vs System 2 (Dual-Process Theory)]]
- [[Cognition (Neuroscience)]]

Constraints:

- Filter all figurative language through Economic and Computer metaphor/analogy.
- Entirely written on remarkable 2 (No AI Paraphrasing).
- Mention every concept from logs 1-6 in the essay in proper context.